

Numerical analysis of PEM water electrolyzer for hydrogen production: Critical parameters

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Abstract

Concerns about the effects of conventional energy sources on the environment have grown in recent years. It is crucial to switch to sustainable renewable energy sources for various uses to address the challenges of depletion and environmental degradation. The use of polymer electrolyte membrane (PEM) electrolyzers for hydrogen production is one of the intriguing solutions that have a lot of potential. Incorporating basic thermodynamics and electrochemical relationships, this chapter gives the mathematical equations and models that relate to PEM electrolyzers. The research takes into account the well-known Butler-Volmer kinetics for the electrodes and transport resistance within the polymer electrolyte, and it concentrates on a simplified PEM electrolyzer. The investigation looks at how temperature affects the ionic conductivity, resistance, and operational cell voltage of the polymer electrolyte. Various exchange current densities are also included. Specifically, the resistance decreases from $0.438 \Omega/\text{cm}^2$ at 298 K to $0.161 \Omega/\text{cm}^2$ at 373 K. Conversely, a slight increase in ionic conductivity is observed. This observation suggests that the membrane assembly's ionic conductivity improves with temperature, which consequently enhances the hydrogen flow rate.

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